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1. The saliency maps highlight the pixels that most influence the model's prediction. In all five examples, the brightest regions roughly correspond to the main object in the image.

   It indicates that the model relies on relevant visual cues for its predictions. This confirms that gradient-based saliency provides a useful, intuitive way to visualize pixel importance.

2. Saliency maps have some limits:

(1) They are highly noisy and unstable, which makes intepretation difficult.

(2) They lack spatial coherence

(3) The method relies on a local linear approximation, which is not accurate for deep nonlinear networks.

(4) Gradients may ~~sta~~ saturate or be dominated by noise, giving misleading importance

(5) The results are sensitive to preprocessing and pixel scaling.

3. Yes. Beides interpretability, saliency can be used for:

(1) Weak object localization, to find approximate object regions without annotations.

(2) Data and label debugging, by spotting which regions influence wrong predictions.

(3) Detecting spurious correlations, such as background biases.

(4) Robustness analysis, for understanding how inputs affect the model.

4. The saliency maps obtained with VGG 16 show clearer and more localized activations on the target objects than those produced by SqueezeNet.

   The deeper architecture allows gradients to propagate in a more structured way, which leads to more interpretable saliency patterns.

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However, the inherent noise and instability of gradient-based saliency remain visible, which indicates that the technique's weakness are largely independent of the chosen network.

5. The fooling image is visually almost indistinguishable from the original quail image, but the model now predicts the target class stingray with high confidence. This shows that the adversarial perturbation doesn't affect human perception but strongly affects the model.

In the raw difference image, the perturbation appears nearly uniform and invisible, while the magnified ~~re~~ version reveals structured high-frequency noise mainly around the bird's body and edges.

These regions correspond to pixels that the model relies on most, meaning that altering them slightly is enough to shift the decision boundary.

Overall, the results illustrate how fragile the network can be: extremely small, carefully crafted changes can completely flip the prediction without altering the semantic content of the image.

6. Adversarial vulnerability has important practical implications:

CNNs can be easily fooled in safety-critical systems.

It shows that models may rely on brittle, non-robust features rather than meaningful semantics, which reduces trust and reliability in real-world deployments.

It also exposes potential security risks, since an attacker can intentionally craft such perturbations.

7. This gradient-ascent has limitations:

(1) It requires many iterations, may produce unstable updates, and often generates unrealistic or high-frequency noise.

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(2) It is also sensitive to step size and can get stuck or overshoot.

More advanced approaches:

FGSM (Fast Gradient Sign Method)

PGD (Projected Gradient Descent)

momentum-based attacks. and optimization-based attacks like C & W

These methods are more effective, more stable.

8. The visualization gradually evolves from random noise into a structured pattern that clearly reflects features of the target class Yorkshire terrier.

After a few iterations, the image begins to show characteristic textures.

As iteration increases, the visualization becomes much sharper and filled with repeated dog motifs, which indicates strong activation of the class-specific neurons.

The final image does not resemble a natural photograph, but instead shows highly colorful, repetitive, high-frequency textures.

This demonstrates that the network relies on complex internal features and local patterns rather than global object shapes.

It also reveals how the model internally represents the concept of a Yorkshire terrier: as a combination of fur-like textures, edges and distinctive facial components.

9. • Iteration: With more iterations, the patterns become sharper and more class-specific, while too few iterations lead to very fuzzy images.

• learning rate: A small learning rate produces slow but stable changes; a very large learning rate can introduce artifacts and make the image unstable.

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- L2 weight: Decreasing the L2 weight allows high-frequency, saturated textures to appear, while increasing L2 yields smoother but less detailed images.

10. Using a real ImageNet image as the source changes the behaviour of class visualization.
Instead of creating a pattern from pure noise, the optimization starts from a natural example of the target class and enhances it.
The final image keeps the global structure of the original picture, but the discriminative features are strongly amplified.

This has two main interests:

(1) Convergence is faster and more stable, because the image is already close to a good solution

(2) It provides an intuitive view of which parts of a real object the network wants to emphasize, which makes the visualization easier to interpret than starting from random noise.

11. Using VGG16 for class visualization leads to more detailed and stable patterns compared to SqueezeNet.
The deeper architecture and higher representational capacity of VGG16 produce gradients that better align with meaningful class features, which yields ~~to~~ clearer textures and more structured activations.
However, the images still exhibit strong high-frequency artifacts and repetitive ~~mod~~ motifs, which indicates that the network relies predominantly on texture-based cues.
Thus, while architecture affects the quality of the visualization, the fundamental behaviour of gradient-based class visualization remains consistent across CNNs.

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